

08 Logistic Regression Insurance

Markdown Cell 1

Experiment 8 — Logistic Regression using Insurance Data
Source: Logistic Regression PDF

Implementation of Logistic Regression using insurance data, including a comparison with Linear Regression.

Code Cell 2

```
import pandas as pd
import matplotlib.pyplot as plt

# Read data
df = pd.read_csv("insurance_data.csv")
df.head()
```

Code Cell 3

```
# Scatter plot
plt.scatter(df.age, df.bought_insurance, marker='+', color='blue')
plt.xlabel("Age")
plt.ylabel("Bought Insurance")
plt.show()
```

Code Cell 4

```
# Train-test split
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    df[['age']],
    df.bought_insurance,
    test_size=0.20,
    shuffle=False
)
```

Code Cell 5

```
# Logistic Regression
from sklearn.linear_model import LogisticRegression

model = LogisticRegression()
model.fit(X_train, y_train)
```

Code Cell 6

```
# Predictions
model.predict(X_test)
```

Code Cell 7

```
model.predict([[70]])
```

Code Cell 8

```
model.score(X_test, y_test)
```

Markdown Cell 9

Linear Regression comparison

Code Cell 10

```
from sklearn.linear_model import LinearRegression

model1 = LinearRegression()
model1.fit(X_train, y_train)

model1.predict(X_test)
```

Code Cell 11

```
model1.predict([[70]])
```

Code Cell 12

```
model1.score(X_test, y_test)
```